

Network Cable Types and Tools

Task 1: Ethernet vs. Fiber Optic Cables (3 Differences & Real-World Use)

Ethernet Cables vs. Fiber Optic Cables

Task 1: 3 Key Technical Differences & Real-World Application Examples

Difference Factor	Ethernet Cable (Cat6 Copper)	Fiber Optic Cable (Optical Glass)
1. Transmission Medium & Signals	Electrical Signals over Copper Uses 4 twisted pairs of insulated copper wires that carry electrical voltage pulses. Vulnerable to electromagnetic interference (EMI) and radio frequency interference (RFI).	Light Pulses over Glass / Plastic Transmits data as pulses of light (laser or LED) through ultra-thin glass or silica fibers. Completely immune to electromagnetic interference (EMI) and lightning strikes.
2. Maximum Range & Distance Limit	Short Distance: Up to 100 Meters Strict standard maximum distance limit of 100 meters (328 ft). Beyond this range, electrical signal attenuation and degradation cause packet loss unless repeaters are added.	Long Distance: Up to 40+ Kilometers Capable of traveling enormous distances with minimal attenuation: Multi-Mode travels up to 550 meters, while Single-Mode easily reaches 10 km to 40 km without repeaters.
3. Bandwidth, Speed & Cost	Moderate Speed, Low Cost Delivers 1 Gbps to 10 Gbps speeds. Highly cost-effective cable rolls and inexpensive termination tools (RJ45 crimper) make installation simple and budget-friendly.	Ultra-High Speed, Higher Cost Supports massive throughput from 10 Gbps to 100 Gbps, 400 Gbps+. Higher initial investment for optical transceivers and specialized termination tools (fusion splicers).

REAL-WORLD APPLICATION EXAMPLES

Ethernet Cable Example: Home Wi-Fi Router & Office LAN

Used to connect desktop PCs, gaming consoles, smart TVs, and printers directly to a home Wi-Fi router or office switch. It provides stable, low-latency, plug-and-play wired connections across rooms under 100 meters at very low cost.

Fiber Optic Example: College Campus Backbone & Data Centers

Used as the high-speed backbone linking separate academic buildings across a sprawling college campus or interconnecting thousands of servers in cloud data centers, delivering multi-gigabit throughput across kilometers without interference.

Task 2: Straight-Through Ethernet Cable Wiring (T568B Standard)

- **Goal:** Show the pin numbers (1 to 8) and color code for the T568B standard:
 - **Pin 1:** White-Orange
 - **Pin 2:** Orange
 - **Pin 3:** White-Green
 - **Pin 4:** Blue
 - **Pin 5:** White-Blue
 - **Pin 6:** Green
 - **Pin 7:** White-Brown
 - **Pin 8:** Brown
- *Note:* For a straight-through cable, both ends of the cable use this exact same pin order.

Straight-Through Ethernet Cable: T568B Wiring Standard

Day 4 Task 2: Pinout Diagram, Wire Color Codes (Pins 1-8), and Termination Rules

RJ45 CONNECTOR & T568B WIRE PINOUT DIAGRAM



STRAIGHT-THROUGH CABLE RULE:

- Both End A and End B must be terminated using the EXACT same T568B color pinout.
- Used to connect DIFFERENT devices: Computer to Switch, Router to Switch, or PC to Wall Outlet.
- Gigabit Ethernet uses all 8 wires (all 4 twisted pairs) in bidirectional full-duplex operation.

T568B PINOUT & COLOR SPECIFICATION

PIN	WIRE COLOR (T568B)	SIGNAL / PAIR
1	White / Orange Stripe	Tx+ (Transmit Positive)
2	Solid Orange	Tx- (Transmit Negative)
3	White / Green Stripe	Rx+ (Receive Positive)
4	Solid Blue	Bi-directional / PoE+
5	White / Blue Stripe	Bi-directional / PoE-
6	Solid Green	Rx- (Receive Negative)
7	White / Brown Stripe	Bi-directional / PoE+
8	Solid Brown	Bi-directional / PoE-

CRITICAL EXAM / LAB TIPS:

1. T568B always starts with Orange (White-Orange / Orange).
2. Pin 3 & Pin 6 form the Green pair (split around Blue).
3. Pins 4 & 5 are Blue; Pins 7 & 8 are Brown.
4. Clip orientation: When looking at copper pins facing you, with the plastic clip facing DOWN. Pin 1 is on the far LEFT.

Task 3: Step-by-Step Guide to Crimping a CAT6 Cable

1. **Strip the Jacket:** Use the wire stripper on the crimping tool to remove about 1.5 inches of the outer PVC jacket, taking care not to nick the inner colored wires.
2. **Untwist & Straighten:** Cut away the central plastic separator (spline), untwist the 4 wire pairs, and straighten each wire with your fingers.
3. **Arrange in T568B Order:** Line the wires up flat and parallel in the exact T568B sequence: *White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown*.
4. **Trim the Wires:** Use the cutter on the crimper to cut the wires straight across, leaving about 0.5 inches of exposed wire.
5. **Insert & Crimp:** Slide the wires firmly into the clear RJ45 connector until all 8 wires touch the front pins and the jacket sits under the internal retaining wedge. Insert the plug into the RJ45 crimping cavity and squeeze the handles firmly until it clicks.

Task 4: Gaming Café Setup Scenario (10 PCs for LAN Gaming)

- **Choice of Cable & Tool:**
 - **Cable: Cat6 Ethernet Cable**
 - **Tool: RJ45 Crimping Tool**
- **Justification (3–4 sentences):**
 - Cat6 Ethernet cable is the optimal choice for a 10-PC gaming café because distances between computers and the switch are well under 100 meters, making expensive optical fiber unnecessary.
 - Cat6 easily delivers 1 Gbps to 10 Gbps speeds with ultra-low latency, ensuring smooth, lag-free multiplayer gaming and local file sharing.
 - An RJ45 crimper is far more cost-effective and simpler to use than a precision fiber cleaver and fusion splicer, allowing fast, durable on-site cable repairs within a modest budget.